

Workshop: Explainable AI (XAI)

Mastering Explainability

2026.04.30

Seung Hyun Han
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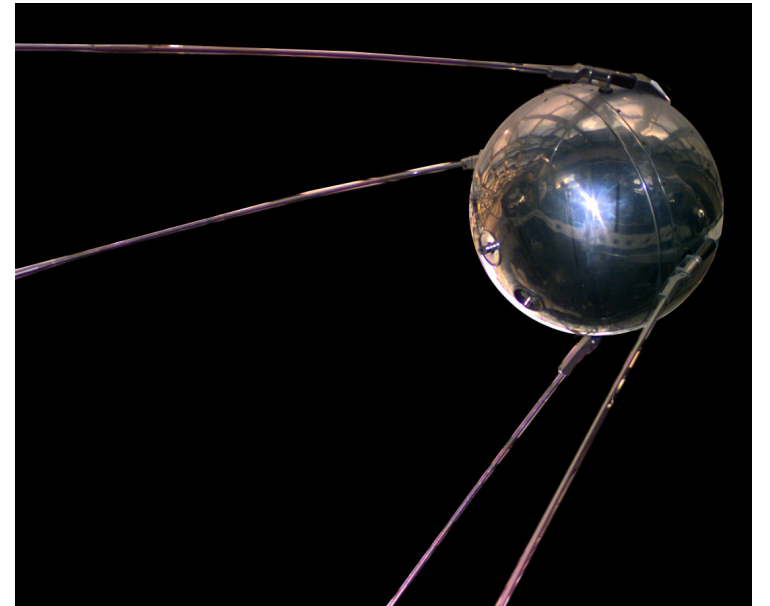
- Preliminary Knowledge
 - Understanding Machine Learning
 - Black-Box Model
 - Visualization vs XAI
- Decision Tree
 - Visualizing Decision Tree
 - Feature Importance
 - Partial Dependence Plot
 - Xgboost
- Surrogate Analysis
 - LIME
 - SHAP

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Introduction

- DARPA (Defence Advanced Research Projects Agency)
 - Sputnik Crisis (Спутниковый кризис) 1957.10.4



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- Dwight David Eisenhower → James Rhyne Killian (former MIT president)
- ARPA (Advanced Research Projects Agency) → "D"ARPA

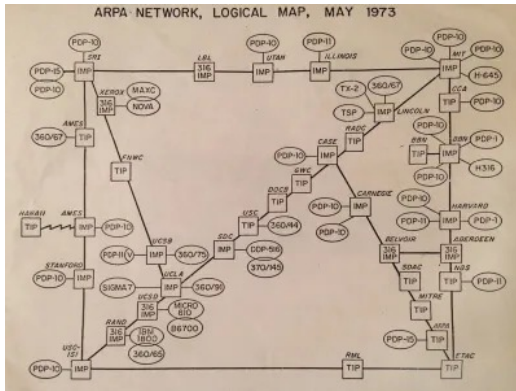
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- ARPnet(1969)

[www based internet](http://www.basedinternet.com) (1990.12.20)

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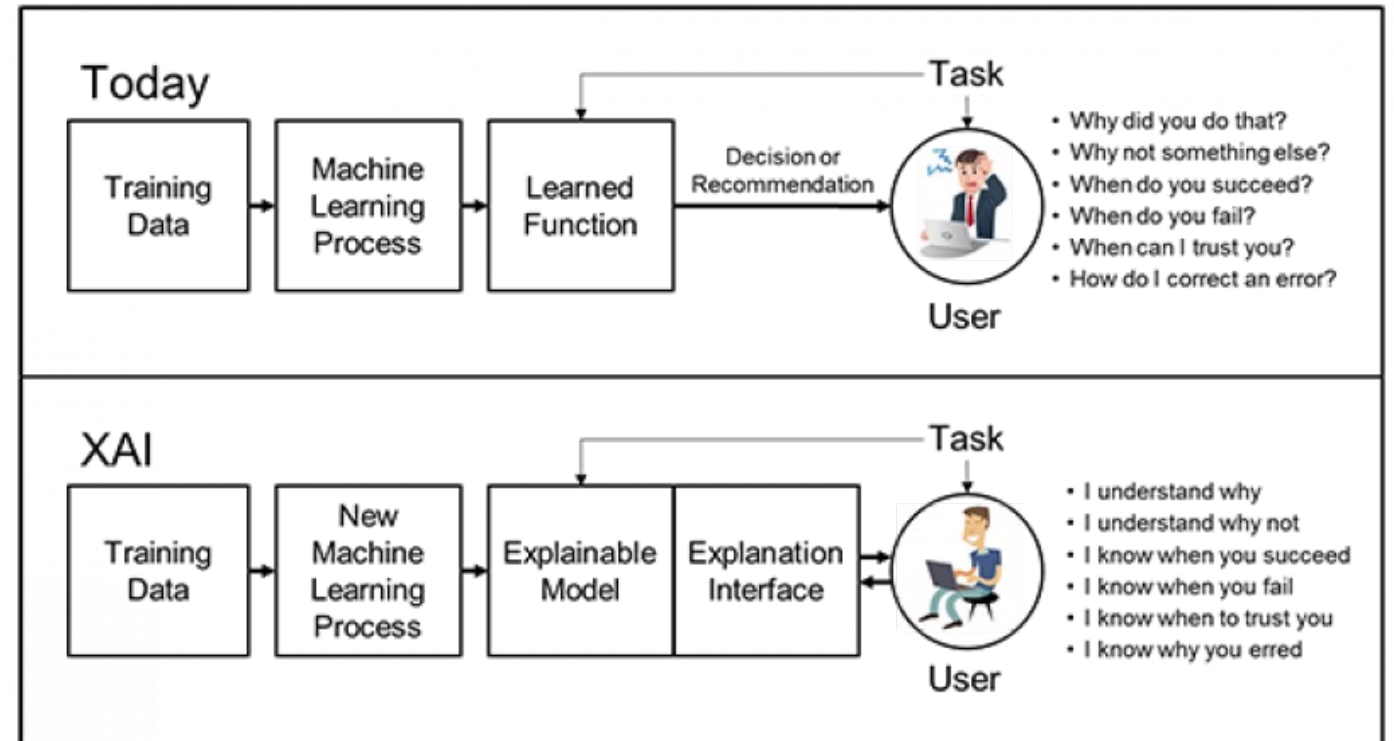
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- 2016.8.10
 - DARPA-BAA-16-53

Broad Agency Announcement
Explainable Artificial Intelligence (XAI)
DARPA-BAA-16-53
August 10, 2016

Google DeepMind Challenge Match	
Google DeepMind Challenge Match	
 Google DeepMind Challenge Match	
period	2016 March 9th - March 15th
host	Google
progress place	Four Seasons Hotel Seoul (Republic of Korea , Seoul Metropolitan City Jongno- gu)
Participant	AlphaGo vs Lee Sedol 9

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 - 2016~2018.11



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 - Outcome : <https://xaitk.org/>

xaitk-saliency Public
As part of the Explainable AI Toolkit (XAITK), XAITK-Saliency is an open source, explainable AI framework for visual saliency algorithm interfaces and implementations, built for analytics and autonomy applications.
Python ☆ 101 📄 Apache-2.0 🗣️ 28 🔄 4 📄 3 Updated on Feb 11

xaitk-saliency-web-demo Public
Demonstration Web UI to visualize XAITK Saliency functionality.
Python ☆ 18 📄 BSD-3-Clause 🗣️ 10 🔄 3 📄 3 Updated on Mar 14, 2025

xaitk.github.io Public
Forked from [mmistakes/minimal-mistakes](#)
Website that hosts the open-source explainable AI toolkit (XAITK). Built with Minimal Mistakes in Jekyll
HTML ☆ 7 🗣️ 27,460 🔄 0 📄 0 Updated on Jan 21, 2025

xaitk-saliency-web-api Public
Web service(s) pertaining to functionality exposed by the xaitk-saliency package.
Python ☆ 1 📄 MIT 🗣️ 3 🔄 1 📄 1 Updated on May 17, 2023

meta Public
Meta repository for capturing issues not strictly relating to another repository here.
☆ 0 🗣️ 0 🔄 0 📄 0 Updated on Apr 10, 2021

Introduction

- XAI (2026–2021) eXplainable Artificial Intelligence
 - Shortliffe, Edward H., and Bruce G. Buchanan. "A model of inexact reasoning in medicine." *Mathematical biosciences* 23.3–4 (1975): 351–379.

ABSTRACT

Medical science often suffers from having so few data and so much imperfect knowledge that a rigorous probabilistic analysis, the ideal standard by which to judge the rationality of a physician's decision, is seldom possible. Physicians nevertheless seem to have developed an ill-defined mechanism for reaching decisions despite a lack of formal knowledge regarding the interrelationships of all the variables that they are considering. This report proposes a quantification scheme which attempts to model the inexact reasoning processes of medical experts. The numerical conventions provide what is essentially an approximation to conditional probability, but offer advantages over Bayesian analysis when they are utilized in a rule-based computer diagnostic system. One such system, a clinical consultation program named MYCIN, is described in the context of the proposed model of inexact reasoning.

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 - Shortliffe, Edward H., and Bruce G. Buchanan. "A model of inexact reasoning in medicine." *Mathematical biosciences* 23.3–4 (1975): 351–379.
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■ Abstract

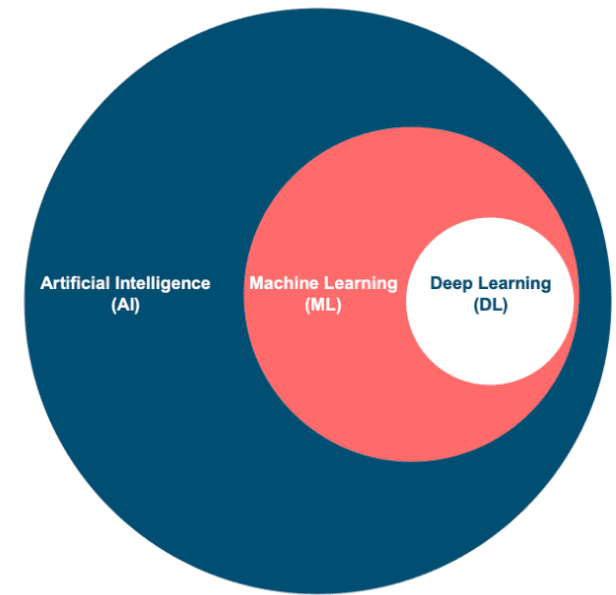
As the artificial intelligence (AI) systems in military simulations and computer games become more complex, their actions become increasingly difficult for users to understand. Expert systems for medical diagnosis have addressed this challenge through the addition of explanation generation systems that explain a system's internal processes. This paper describes the AI architecture and associated explanation capability used by Full Spectrum Command, a training system developed for the U.S. Army by commercial game developers and academic researchers.

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Preliminary Knowledge : Re-visiting Machine Learning

- ML, DL and AI
 - AI: Capability of computer systems to perform tasks that typically require human intelligence.
 - **ML**: Algorithms learn patterns from data to make predictions or decisions without being explicitly programmed for each task.
 - DL: A multilayered neural networks to automatically learn hierarchical representations from large amounts of data.



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- ML vs DL
 - Data management, or so called, **Feature management**
 - Feature?

	Pclass	Age	Survived	Sex_male	Embarked_Q	Embarked_S	family
0	3	22	0	1	0	1	1
1	1	38	1	0	0	0	1
2	3	26	1	0	0	1	0
3	1	35	1	0	0	1	1
4	3	35	0	1	0	1	0

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 - A measurable non-target attribute(property)
 - Domain Knowledge

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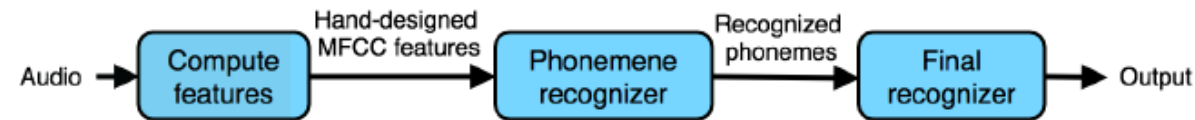
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 - A measurable non-target attribute(property)
 - Domain Knowledge
 - the specialized expertise and understanding of the specific field or subject

	Pclass	Age	Survived	Sex_male	Embarked_Q	Embarked_S	family
0							
1	Titanic - Machine Learning from Disaster Start here! Predict survival on the Titanic and get familiar with ML basics						
2	-	--	-	-	-	-	-
3	1	35	1	0	0	1	1
4	3	35	0	1	0	1	0

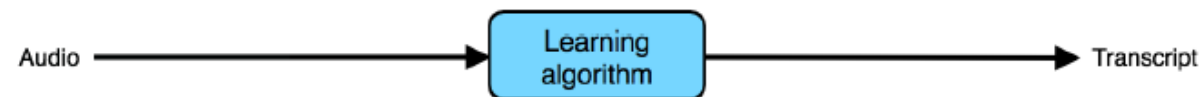
Preliminary Knowledge : Re-visiting Machine Learning

- What is the Model?
 - ML/DL model is a mathematical or computational program that has learned **patterns** from **data** so it can make **predictions** or **decisions** on new, unseen inputs.
- DL with End-to-end?

Traditional model



End-to-end learning



This works well given enough labeled (audio, transcript) data.

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- Black-box model?



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 - In social science, an “interpretability” is...
 - A process whereby a theory is transformed to a level that is understandable to its domain

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 - In social science, an “interpretability” is...
 - A process whereby a theory is transformed to a level that is understandable to its domain
- Do you remember the previous lecture?
 - 3 major categories in contemporary research in AI?
 - Domain adaptation
 - Model lightening
 - XAI

Black Box Model

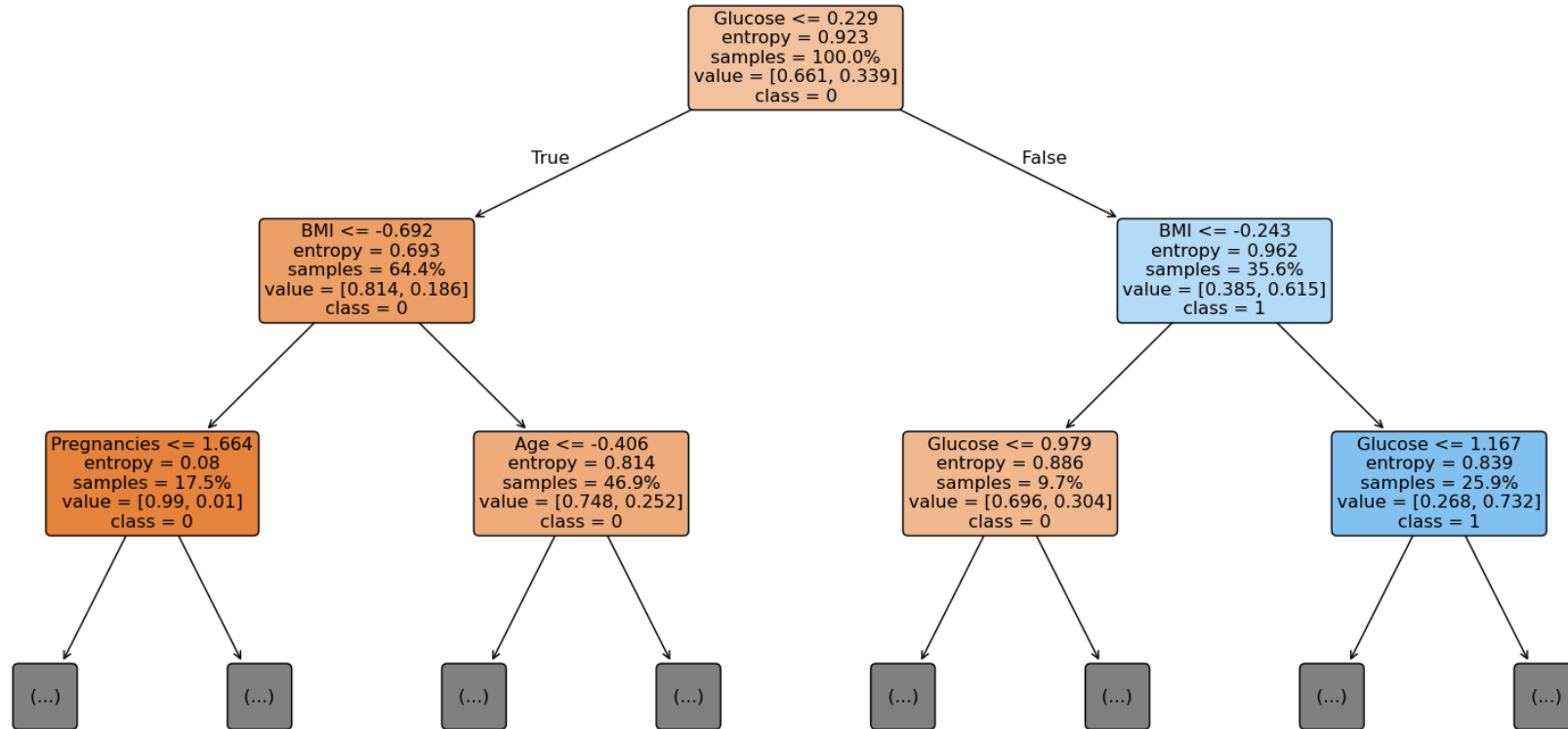
- Case study: diabetes issue in Pima Indians
 - Doctors collected native Pimas blood information

	id	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	576,000000	576,000000	576,000000	576,000000	576,000000	576,000000	576,000000	576,000000	576,000000	576,000000
mean	385,189236	3,866319	120,170139	68,421875	21,067708	82,133681	31,843403	0,479325	33,354167	0,338542
std	221,420334	3,388140	32,014163	20,080641	15,933359	115,444040	7,798820	0,343584	11,972995	0,473625
min	0,000000	0,000000	0,000000	0,000000	0,000000	0,000000	0,000000	0,078000	21,000000	0,000000
25%	192,750000	1,000000	99,000000	62,000000	0,000000	0,000000	27,300000	0,244000	24,000000	0,000000
50%	385,500000	3,000000	116,000000	72,000000	23,000000	45,000000	32,050000	0,369000	29,000000	0,000000
75%	575,250000	6,000000	140,000000	80,000000	33,000000	130,000000	36,600000	0,637750	41,000000	1,000000
max	766,000000	17,000000	199,000000	114,000000	99,000000	846,000000	59,400000	2,420000	81,000000	1,000000

- Draw a Tree

Black Box Model

- Case study: diabetes issue in Pima Indians
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 - Draw a Tree



Black Box Model

- Case study: diabetes issue in Pima Indians
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First record from test dataset:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age
0	-0.846722	2.276904	-0.918192	-0.318333	2.539073	-0.762752	0.511747	-0.781951



First record from original (unscaled) test dataset:

	id	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age
0	258	1	193	50	16	375	25.9	0.655	24

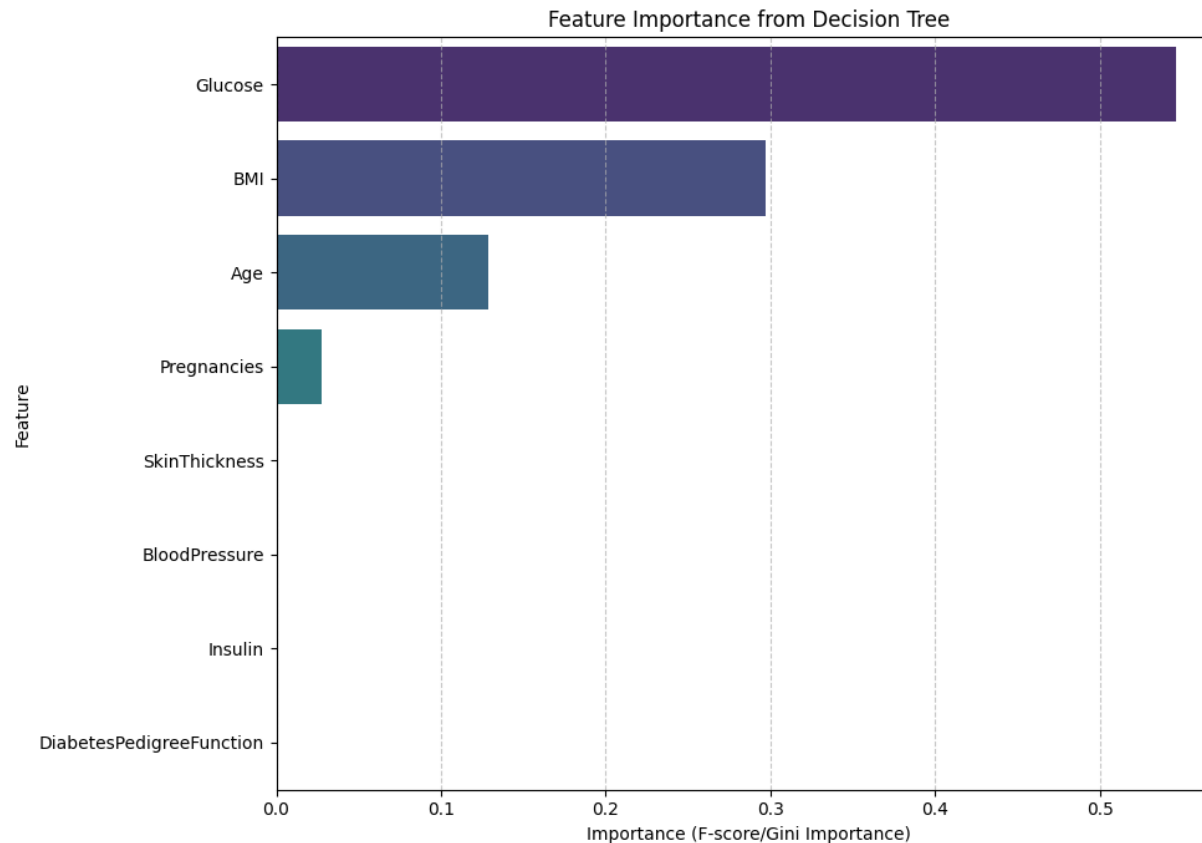


Leaf node class counts (Class 0: 0.3889, Class 1: 0.6111)

Probability of Diabetes for this record: 0.6111

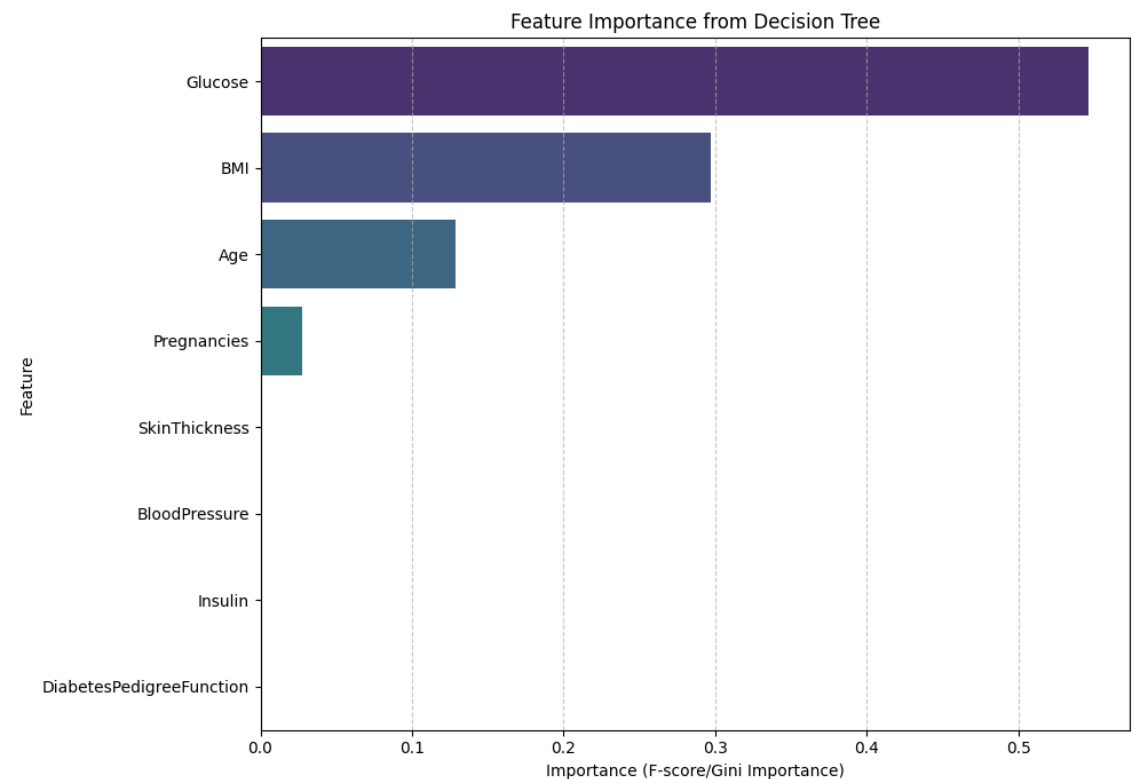
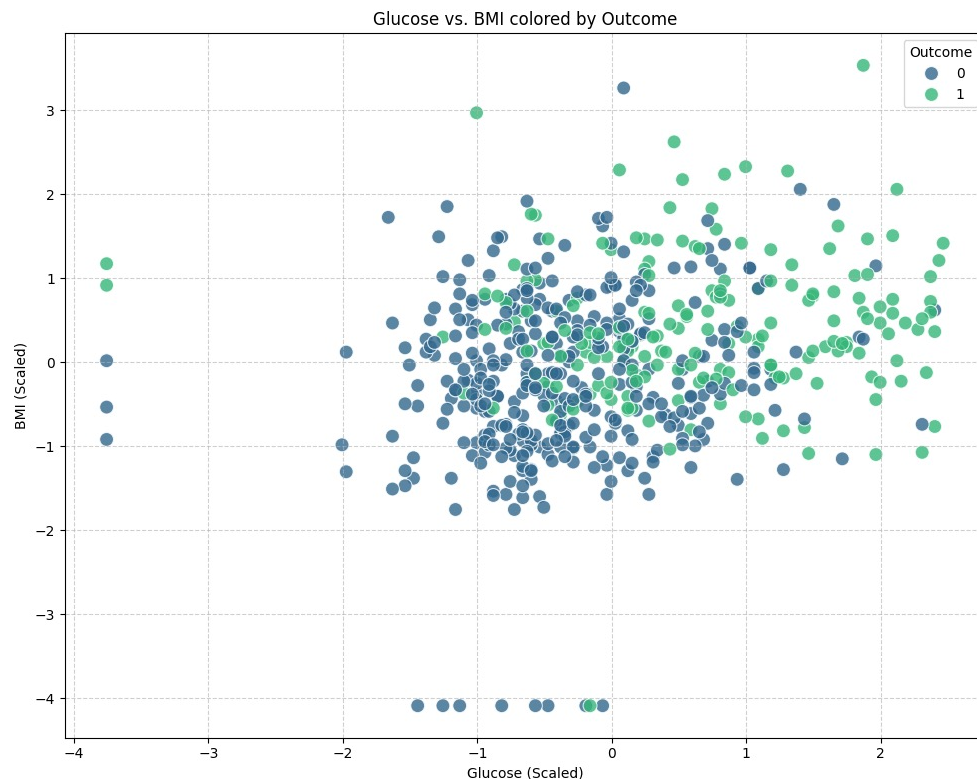
Black Box Model

- Case study: diabetes issue in Pima Indians
 - Doctors collected native Pimas blood information
 - Draw a Tree
 - Draw a Feature Importance diagram



Visualization vs XAI

- Now it comes to have a question like...
 - Isn't it a just visualization for certain models?
 - No, It looks like, but as long as the “interpretability” defines a ‘feature’s’ impact, it’s different from data visualization.
 - Back to Pima cases
 - Visualization & Feature Importance graph



Visualization vs XAI

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 - Isn't it a just visualization for certain models?
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 - Back to Pima cases
 - Visualization & Feature Importance graph
 - Visualization
 - Tools such as histograms, scatter plots, and line graphs that show distributions, trends, and correlations in the data.
 - Understand the data
 - XAI
 - explain how each input factor contributed to a specific prediction and in which direction
 - Understand the model’s decision-making logic

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Fin.

Thank you.

* 이 PPT는 아주대학교에서 제공한 아주체과 아모레퍼시픽의 아리따부리, consolas 글꼴를 사용하고 있습니다.